ANOVA

Which features differ somewhere across three or more conditions, and which conditions those are.

AT A GLANCE

Where	Statistics → ANOVA
Data levels	Protein, phosphosite, phosphoprotein
Needs	At least two conditions with at least two samples each. The post-hoc needs at least three conditions.
The test	One-way ANOVA per feature, with the condition as the single factor and every condition in the annotation as a level.
Two sources	ANOVA draws the significant features; ANOVA Cluster additionally cuts them into groups that move together.
Post-hoc	Tukey HSD, optional, run only on the features the ANOVA called significant.
Tables	The result table, the selected cluster's rows, and the post-hoc comparisons — each downloadable as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What the plot answers

A volcano compares two conditions. With four there are six pairs, and testing all six separately answers six questions when the question asked was one — does this feature differ anywhere across the design — and gives six chances to clear α by accident.

The ANOVA asks that question directly. For each feature it compares the variation *between* the condition means with the variation *within* them, and the p-value is the probability of a ratio at least that large if all the condition means were equal. One test per feature, one correction across features, and a result that says *something moved* without saying where. The figure is a heatmap of the features that passed, so the shape of the movement is visible even though the omnibus test does not name it.

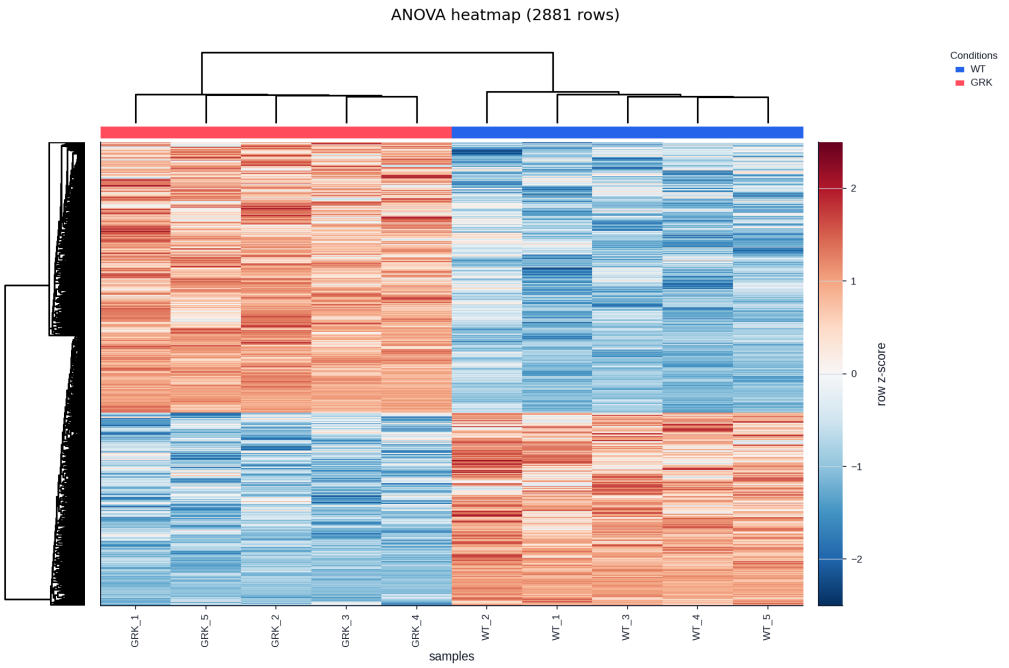


Figure 1.1. The ANOVA heatmap on the example data, at protein level with its two conditions: the 2,881 rows its title counts, against the ten sample columns, each cell a row z-score on the fixed −2.5 to +2.5 scale. The strip above the columns is the condition, and the column dendrogram separates the five GRK samples from the five WT ones without having been told which is which.

Reading it

- One cell: one feature in one sample as a row z-score — how far that sample sits from the feature's own mean, in the feature's own standard deviations across samples. Blue is below, red above; a missing value is grey.
- One row: a feature the ANOVA called significant. Rows are not labelled — there are usually too many — so a feature is found in the table rather than on the figure.
- One column: one sample, labelled condition_n.
- The strip above the columns: each sample's condition, in the colours every other plot in the session uses.
- The dendrograms, when **Show dendrograms** is on: average linkage on euclidean distances, over the z-scored rows and over the samples. In ANOVA Cluster the row tree is replaced by a strip of cluster colours.
- The colourbar, row z-score, fixed from −2.5 to +2.5 so two heatmaps can be compared.

The heatmap visualises features selected by the statistical analysis. Condition separation here is not independent confirmation of that analysis: the rows were selected because their means differ. A row z-score describes relative position within one feature, not an absolute intensity or a fold change shared across rows. For values 8, 9 and 10, centring produces -1, 0 and 1 before division by the row standard deviation; a much more abundant feature with the same relative pattern can have the same colours. Inspect original replicate values and post-hoc results when following up a cluster. Lack of a neat band is not, by itself, a reason to change the significance threshold.

> **EASY TO GET WRONG**
>
> A significant ANOVA says the condition means are not all equal. It does not say which condition differs, or in which direction, and the heatmap is not evidence for a particular pair either — the eye will find one. Set **Post-hoc test** to Tukey HSD to get the pairwise comparisons with their own p-values, and read the direction from the log2FC column of the post-hoc table rather than from the colours.

2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Protein, phosphosite or phosphoprotein. Only the levels uploaded are offered.	Protein	page
Source	ANOVA draws every feature below the threshold. ANOVA Cluster cuts those features into groups and colours them.	ANOVA	page
Data Source	The matrix the test runs on: Data, Filtered Data or Imputed Data, whichever exist.	the most processed matrix available	page
Label column	Which column names a feature in the tables: the workflow name or the gene names.	the workflow name	page
P-value threshold	The cutoff for significance, and the gate on both the heatmap and the post-hoc.	0.05	page
Use uncorrected p-values	Uses the raw p-value in place of the adjusted one everywhere it is read, including the two gates.	off	page
Show dendrograms	Draws the row and column trees beside the heatmap. Turning them off does not change the ordering, which is computed either way.	on	page

Controls on this page (continued)

Control	What it does	Default	Where
Clusters	ANOVA Cluster only. How many row clusters to cut the tree into.	4	page
Cluster method	ANOVA Cluster only. Average (UPGMA), Ward, Complete, Weighted (WPGMA) or Single, always on euclidean distances. Average chains and can return two large clusters and two of a handful of features; Ward is the usual remedy.	Average (UPGMA)	page
Cluster	ANOVA Cluster only. Draws one cluster on its own below the full heatmap and lists its rows. The menu shows each cluster's size.	None	page
Show feature names	ANOVA Cluster only. Writes the **Label column** beside each row of the cluster heatmaps, to the right of the heatmap. The size fits the number of rows, so a single small cluster is readable and a cluster of a thousand rows is not; the plain ANOVA heatmap never names its rows.	off	page
Post-hoc test	None or Tukey HSD.	None	page
Condition 1, Condition 2	In the **Post-hoc Comparison** card. Restricts which pair is *reported*; the numbers are unchanged, because the studentized range still takes k from the whole design and the pooled error from all of it.	the first two conditions in metadata order	page

In the plot appearance panel

Control	What it does	Default	Where
Condition colours, condition labels	The strip above the columns and the names in its legend, shared with every other plot in the session.	the condition palette	appearance
Legend	Hides the cluster and condition legends beside the heatmap.	shown	appearance
Row labels	Size and colour of the feature names. Offered only while **Show feature names** is on. An empty size fits the names to the rows.	fitted, black	appearance
Title, axis labels, font, frame, grid, background	As for every other plot; described in Part XI.	the plot's own	appearance

WORTH KNOWING

The appearance panel opens with **Ctrl** + **Shift** + **E**. It offers no colour-scale card here: the heatmap gradient is fixed at RdBu reversed between −2.5 and +2.5 so that two ANOVA figures can be compared. The figure is also rendered at a fixed size, with no width, height or resolution controls on this page.

3 How it is computed

The ordinary one-way ANOVA assumes independent observations and a common within-condition variance; its small-sample reference distribution also relies on approximately normal errors. It is not a repeated-measures or Welch ANOVA. Check the experimental design before choosing it. A pooled intensity Q–Q plot does not test these assumptions for every feature. Imputed values enter the calculation as values, without propagating their imputation uncertainty.

THE TEST

The factor is the condition, and its levels are *every* condition in the annotation — there is no condition selector on this page, so a condition is excluded from the ANOVA by excluding its samples from the annotation. Each condition must contribute at least two sample columns, and each feature must have at least two observed values in every condition or its row is dropped before the test.

For a feature with k conditions and N observed values in total, the statistic is the ratio of the variation between the condition means to the variation within them:

$$F = \frac{\text{between-condition mean square}}{\text{within-condition mean square}}, \qquad df = (k-1,\ N-k),$$

and the p-value is the upper tail of that F distribution. Missing values are omitted rather than imputed, so N and the within-condition counts are per feature.

A feature whose values are identical in every condition has no defined F. Such rows return no finite p-value and are removed *before* the correction, so they do not sit in the Benjamini–Hochberg denominator. A feature that is constant within every condition but differs between them is different: its F is infinite (or extremely large) and its p-value 0; it is kept, marked in the zero_variance column, and a warning says how many such rows there are, because they rank first for absent variance rather than a large difference.

The page states how many rows were not tested, in two warnings: u of T rows were not tested: fewer than 2 values in at least one condition., and v rows were not tested: their values are identical in every condition, so no F statistic exists.

CORRECTION

The p-values of the surviving features are adjusted by the Benjamini–Hochberg procedure, exactly as on the volcano page: sorted ascending, scaled by m/i, made monotonic from the largest rank downwards and capped at 1, with m the number of features actually tested. adj_pval retains that adjusted result. **Use uncorrected p-values** puts the raw value in selected_pval; the correction column records the selection. Downstream selection — the significance label, which rows reach the heatmap, which rows are clustered, which rows reach the post-hoc — reads that one column.

TUKEY HSD

Run only when **Post-hoc test** is Tukey HSD, only on features whose adjusted p-value is below the threshold, and only when the design has at least three conditions; with two, the omnibus F and the single pairwise comparison test the same hypothesis and the post-hoc is skipped with a notice.

Within a feature the error is pooled across all k conditions as the ANOVA itself pools it, and for a pair a, b:

$$q = \frac{|\bar{x}_a - \bar{x}_b|}{\sqrt{\frac{1}{2}\,\mathrm{MSE}\left(\frac{1}{n_a} + \frac{1}{n_b}\right)}},$$

with the p-value from the studentized range distribution on k and $N - k$. This is the Tukey–Kramer form, so unequal group sizes arising from missing values are handled correctly. The correction is two-layered and deliberately so: Benjamini–Hochberg controls the false discovery rate across features, Tukey the family-wise rate across the condition pairs within each feature.

The reported difference is *signed* and follows the same convention as everywhere else in CoPYlot: the condition named first in the comparison is the minuend, so log2FC is mean(group1) − mean(group2) and a positive value means higher in group1. The q statistic uses the absolute difference, so the p-value itself is symmetric.

CLUSTERING, IN ANOVA CLUSTER

The significant rows are z-scored across samples — each row minus its own mean, divided by its own standard deviation, with a zero standard deviation replaced by 1. Rows not observed in every sample cannot be clustered and are dropped; the count is reported under the plot. The rest are clustered by the chosen linkage on euclidean distances, cut into the requested number of groups, and the groups renumbered down the dendrogram, so cluster 1 is the topmost band of the figure rather than an arbitrary label.

Edge cases

- A condition with one sample column stops the whole ANOVA with a message naming it; it is not run on the remaining conditions.
- When nothing passes the threshold, no heatmap and no clusters are produced and the panel says so.
- Requesting more clusters than there are significant rows gives as many as the tree can supply.
- The heatmap tolerates missing values — drawn grey — but the dendrograms and the clustering do not, so switching either on can drop rows the plain heatmap showed.
- Tukey costs one studentized-range evaluation per feature per pair. Naming a pair is the cheap way to ask, and returns identical numbers.

4 The table below the plot

The result table has one row per feature tested, sorted by adjusted p-value, then raw p-value, then label. It carries the label column, every sample column that went into the test, and:

Columns in this table

Column	What it holds
f_statistic	The ANOVA F for that feature.
pval	The raw p-value from the F distribution.
adj_pval	The Benjamini–Hochberg adjusted p-value.
selected_pval, correction	The raw or adjusted p-value selected for classification, and its stated basis.
significance	Significant or Not significant, against the **P-value threshold**.
neg_log10_selected_pval	$-\log_{10}$ of the value used, clipped at 10^{-300}.
mean_<condition>	One column per condition: the mean of that condition's observed values for the feature.
cluster	ANOVA Cluster only, in the cluster table: which group the row was cut into.

The post-hoc table is separate: one row per feature and condition pair, sorted by Tukey p-value, with a header stating which comparison was run and how many were significant.

Columns in the post-hoc table

Column	What it holds
comparison, group1, group2	The pair, as group1 vs group2 and as its two names, in the order compared.
log2FC	mean(group1) − mean(group2).
n_group1, n_group2	Observed values behind each mean.
q_statistic, tukey_pval	The studentized range statistic and its p-value for this pair. The p-value is not adjusted again across features.
significance	Against the same **P-value threshold**.
anova_adj_pval	The feature's ANOVA adjusted p-value, carried through so the row reads on its own.

5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
anova_df = anova_data_f(data, meta, workflow="Protein")
fig = anova_plot(anova_df, meta)
```

R

```
anova_df <- anova_data_f(data, meta, workflow = "Protein")
fig <- anova_plot(anova_df, meta)
```

As on the volcano page the analysis is two calls: anova_data_f returns the result table and anova_plot draws the rows of it that pass, so the *F* statistics and the adjusted p-values are in hand before any heatmap exists. Leaving conditions unset uses every condition in the annotation, which is what the page does, since it has no condition selector; the two other panels are anova_cluster_data into anova_plot_cluster, and anova_posthoc_data for Tukey. The one visible difference is the key to the condition strip: the strip is drawn either way, but the legend naming its colours — which the page always shows — is behind show_condition_legend, and that is off by default.

6 Exporting

Each figure has its own Open Image link and each table its own Download CSV: the full result table, the selected cluster's rows, and the post-hoc comparisons. The heatmap is a PNG at a fixed size, so a figure for a paper is usually better assembled from the cluster table than cropped out of the full square.

Behind this page

Function	What it produces
anova_data_f	The result table: *F*, p-values, adjusted p-values, per-condition means
anova_cluster_data	The z-scored clustering of the significant rows
anova_posthoc_data	Tukey HSD for every condition pair, or one named pair
anova_plot	The heatmap of the significant features
anova_plot_cluster	The same heatmap with the cluster strip